DOT 1.1.0

DOT - How It Works and How to Use It

A concise guide to the Dipole Optimization Tool

Purpose

DOT automatically synthesizes two-dimensional superconducting dipole cross-sections from electromagnetic targets, cable data, and geometric constraints. It searches for a Pareto set instead of copying a reference layout.

DOT calculates	DOT does not calculate
Block geometry, target-field current, bore harmonics, peak field, short-sample current, load-line margin, and geometric feasibility.	Iron, coil ends, persistent-current effects, mechanics, protection, tolerances, cryogenics, or any three-dimensional behavior.

Correct interpretation: a certified candidate satisfies the declared DOT equations and straight-section constraints during final verification. It is an electromagnetic pre-design, not a construction release.

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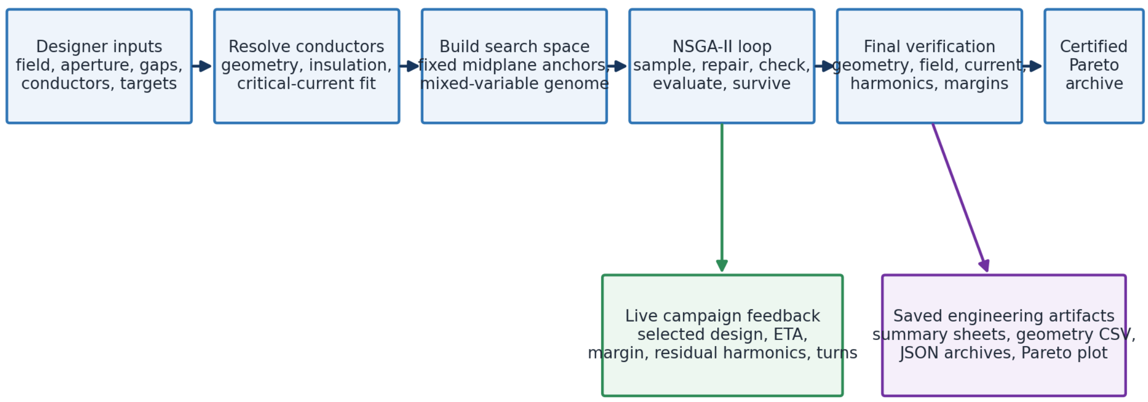
PART I - HOW IT WORKS

One workflow, from requirements to a certified Pareto archive

The optimizer never receives a known coil layout. It receives only the design space, the conductor data, and the required performance.

DOT end-to-end workflow



The optimizer never receives a reference coil layout. It receives only the declared design space and requirements.

Stage	What happens	Why it matters
1. Define	Field, aperture, cables, layers, gaps, block/turn bounds, current and acceptance targets.	These inputs determine which designs can exist.
2. Resolve	Each conductor name is linked through the cadata catalogue to cable, strand, insulation and critical-current fit.	Geometry and load-line physics use one consistent data chain.
3. Search	NSGA-II breeds, repairs, checks and evaluates mixed continuous, integer and binary genomes.	Field quality and margin are explored together without weights.
4. Certify	The final hard-feasible archive is recalculated with fixed final settings and every target is re-applied.	Only this stage can label a candidate certified.
5. Export	DOT saves the Pareto front, representative design, alternative topologies, geometry tables and provenance.	The result can be inspected and reproduced.

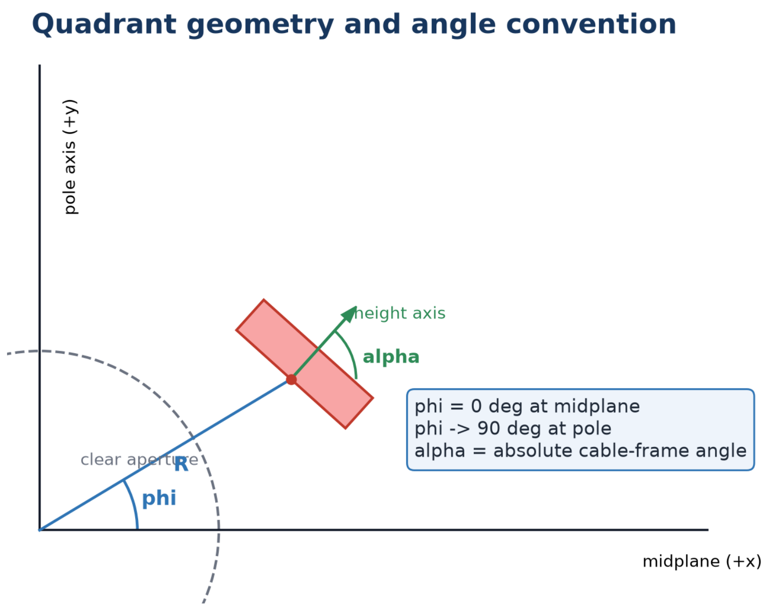
The essential distinction

Search values guide evolution. Final checks decide acceptance. A live best candidate is useful progress information, but it is not a final pass until the end-of-run checks succeed.

PART I - HOW IT WORKS

Geometry: fixed anchors, variable blocks

DOT works in the first quadrant. The midplane is +x, the pole direction is +y, phi starts at zero on the midplane, and alpha is the absolute cable-frame angle.



Layer 1 anchor	Later-layer anchor
R1 = aperture radius phi1 = atan(azimuthal gap / R1) alpha1 = 0 deg	R is the outermost x coordinate of the previous layer's first insulated turn plus the radial gap. Then phi = atan(gap / R), alpha = 0 deg.

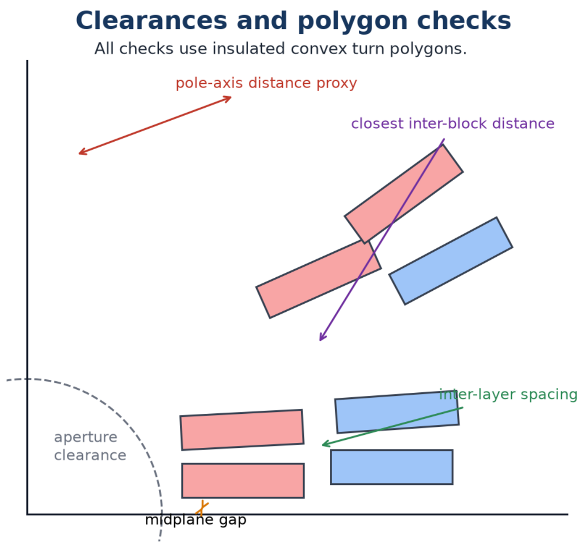
Quantity	Midplane block in every layer	All later block slots
Existence	Always active	Optimizer may activate a contiguous prefix
Layer radius R	Derived from aperture and radial gaps; fixed	Same fixed layer radius
Turns	Integer chosen by optimizer	Integer chosen when active
phi	Derived from azimuthal gap; fixed	Continuous optimizer variable
alpha	Fixed at 0 deg	Continuous optimizer variable

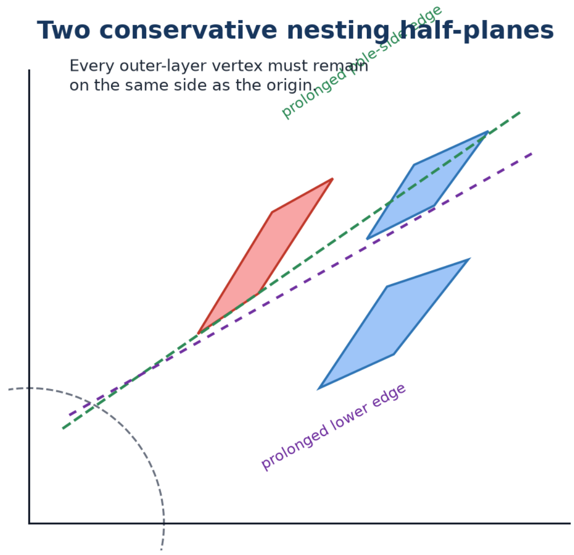
Real geometry: every insulated turn is a convex quadrilateral built from the cable dimensions, keystone, insulation, R, phi and alpha. Turns are arc-stacked from midplane toward pole. Block numbers run continuously across layers, beginning with Layer 1's midplane block.

PART I - HOW IT WORKS

Feasibility is decided on the actual insulated turn polygons

Constructive placement and repair make the search efficient; exact geometric checks remain the ground truth in every generation and again during certification.





Rule	What DOT verifies
Aperture and midplane	No insulated polygon enters the clear bore or crosses the symmetry plane.
Turn collision	No positive-area intersection within a block, between blocks, or between layers.
Inter-block clearance	Exact closest polygon distance in each layer is at least the requested value.
Radial separation	Adjacent layer radii preserve the requested insulated radial build.
Pole-turn proxy	The closest Layer-1 insulated point stays at least the requested distance from the pole axis.
Layer nesting	Every outer-layer vertex remains on the aperture side of both prolonged long-edge lines from the inner pole turn.
Topology and turns	Active blocks are contiguous and all block/layer/total turn limits are respected.

Nesting in plain language

Imagine extending both long sides of the inner layer's pole-most turn as infinite lines. The complete outer layer must remain behind both lines, toward the aperture. This conservative envelope rejects layouts that may fit on paper but are unfavorable to wind as nested layers.

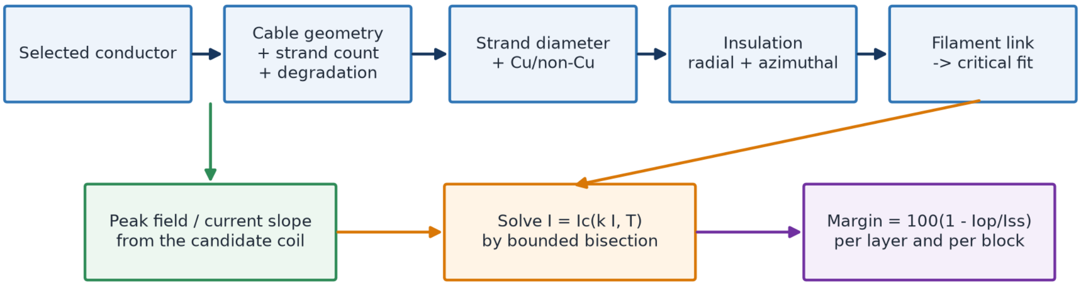
Numerical tolerance is not a design gap. It only classifies round-off and near-contact. Reserve the physical clearance you actually need.

PART I - HOW IT WORKS

Physics: geometry -> field -> harmonics -> load-line margin

The forward engine assumes infinitely long z-directed currents, exact dipole mirror symmetry, linear materials and no iron.

Conductor catalogue resolution and load-line calculation



Calculation	Meaning
Bore field	Biot-Savart fields from distributed line-current sources are summed at the center. In range mode, linearity gives the exact current required for the target field.
Harmonics	Normal coefficients b3, b5, ... are reported in accelerator units at the reference radius: one unit is 1e-4 of the main normal dipole field.
Peak field	DOT samples the bare conductor boundary. The most exposed block/layer controls its load line.
Short-sample current	For each conductor, DOT intersects the geometry's peak-field/current slope with the linked critical-current surface at the selected temperature.
Margin	margin = 100 x (1 - Iop / Iss). DOT reports it per block and layer and optimizes the minimum value.

Current range mode

If Min Current and Max Current differ, DOT scales the series current to the requested bore field, then enforces both bounds.

Fixed-current mode

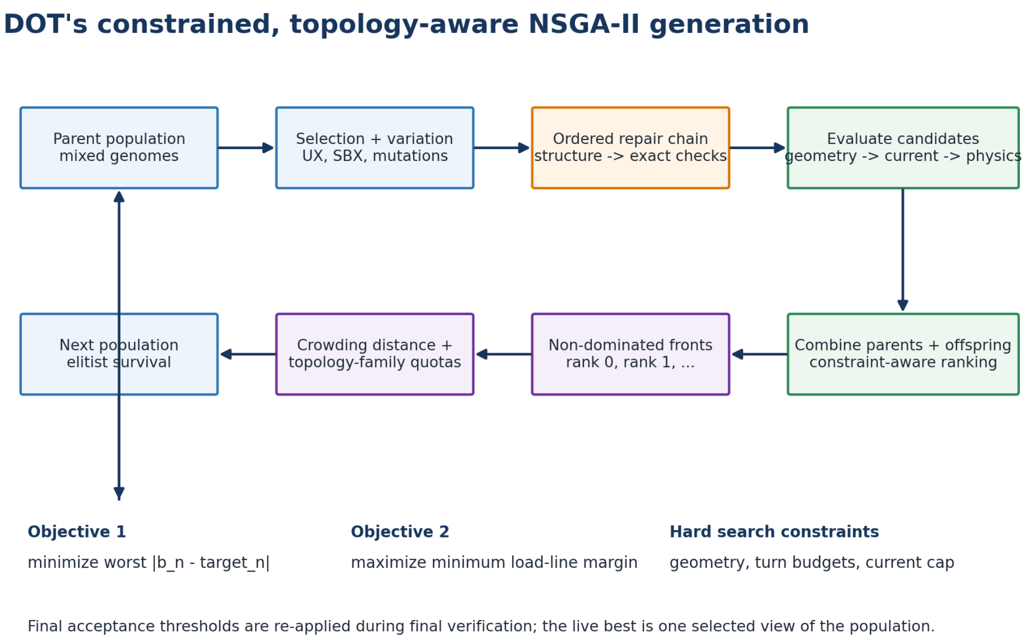
If Min Current equals Max Current, DOT applies that exact current and requires the geometry to reproduce the bore field within 0.01%.

Supported cadata critical surfaces are REMFIT type 1 (Nb-Ti) and type 11 (Nb3Sn). The conductor name must exactly match the selected catalogue record. The GUI's current advice is a conservative Layer-1 sanity check, not a hard limit.

PART I - HOW IT WORKS

NSGA-II keeps the harmonic-margin trade-off visible

DOT has two electromagnetic objectives and no hidden weighted sum: minimize the worst requested harmonic residual and maximize the minimum load-line margin.



Concept	Meaning in DOT
Dominance	A candidate dominates another only if it is no worse in both objectives and better in at least one.
Hard constraints	Geometry, turn budgets and current feasibility take priority over objective quality.
Diversity	Crowding distance spreads survivors along the front; topology-family quotas keep distinct block-count patterns alive.
Variation	Parent selection, crossover and mutation change active blocks, turns, phi and alpha; repair restores ordered, packable winding structure.
Complexity preference	Fewer turns, then fewer blocks, are preferred only when electromagnetic objectives are numerically equivalent.

Final certification

After the last generation DOT deduplicates the hard-feasible archive and recalculates it with fixed final settings (12 x 12 bore and 80 x 80 peak-field midpoint sources per turn), re-applies every target, and rebuilds the non-dominated certified front.

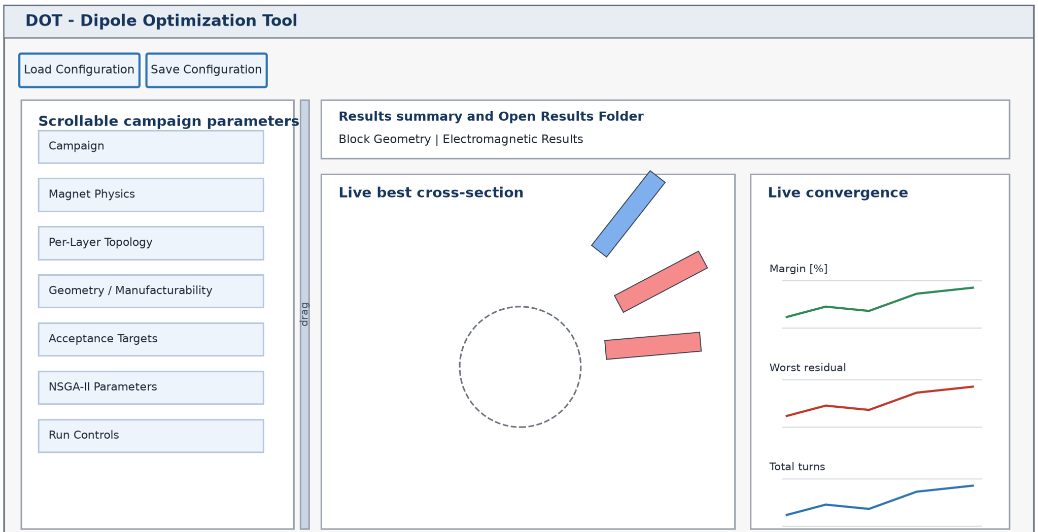
Optional radial-block preference (1.1.0)

Off by default. Target-met layouts enter a separate persistent archive even when electromagnetically dominated. After three reproduction cycles, DOT seeds 5% of offspring from archived elites. Each trial gradually adapts one free block's phi and alpha, preserves turns and topology, and must pass every geometry check while improving radial RMS. Final selection still requires all targets, then compares radiality before surplus harmonic or margin performance.

PART II - HOW TO USE IT

Launch once, then configure the campaign in a physical order

On Windows, double-click launch_dot_gui.cmd. Python is the only prerequisite; the launcher lists dependencies, asks permission, creates .venv, installs them, and opens DOT.



Left: define the campaign. Right: inspect the best current design, detailed tables, and three generation trends.

Order	Set	Good practice
1	Campaign name and output directory	Use a new descriptive name; every run receives a timestamped folder.
2	Field, aperture, temperature, reference radius, layers	Start with Rref = round(2/3 aperture radius, 3); change it only for a defined field-quality convention.
3	One cadata file and supported conductor per layer	Use the selector; the conductor name must match the file exactly.
4	Min/Max blocks and Min/Max Turns	Keep enough independent blocks for harmonic control without forcing unnecessary topology.
5	Azimuthal/radial gaps, block clearance, pole proxy, nesting	Enter physical manufacturing clearances; Layer 1 has no radial-gap input.
6	Harmonic targets, margin and current bounds	Signed harmonic targets can pre-compensate a separately calculated yoke contribution.
7	Search effort, seed and optional parallel evaluation	Use a fixed seed for reproducibility; test parallel mode on the actual hardware.

Quick exploration
160 population x 80 generations. Check reachability and input mistakes.

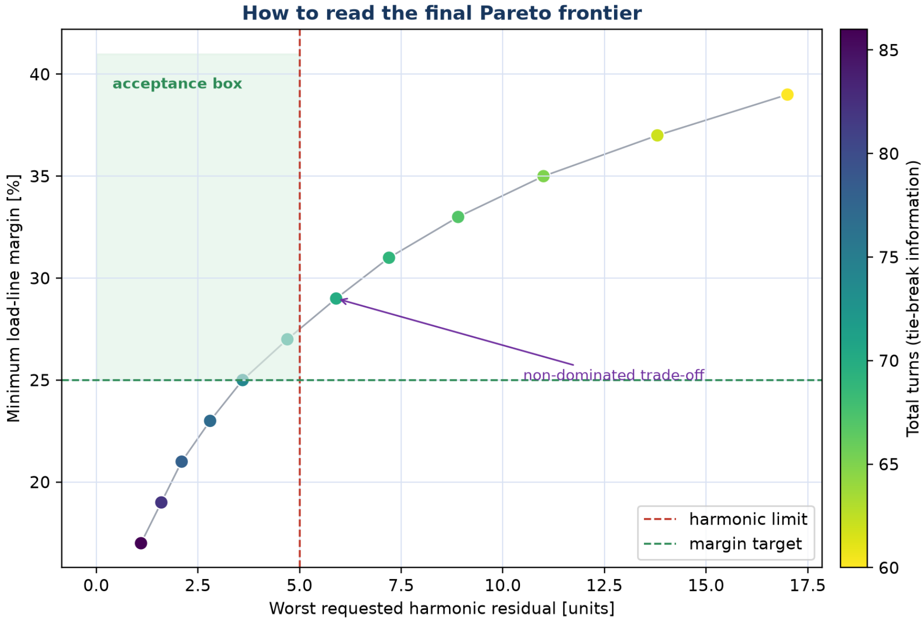
Design search
480 x 200. Recommended normal synthesis campaign.

Intensive search
1000 x 300. Broad, slow search for demanding multilayer designs.

PART II - HOW TO USE IT

Follow progress, then trust the saved certified result

Press Start Campaign only after reviewing the current form. DOT snapshots those values, clears the previous plots, and uses the new targets immediately.



During the run

Watch generation, ETA, live cross-section, every requested harmonic, every block margin, and the margin/residual/total-turn trends.

At the end

A certified point must lie inside the acceptance box after dense re-evaluation. Select a point with reserve, not one exactly on every limit.

Saved artifact	Use it for
best_candidate_summary.png	One-page visual: cross-section, electromagnetic results and block geometry.
best_candidate.json	Complete machine-readable certified representative.
best_candidate_geometry.csv	Per-block R, turns, phi and alpha for reconstruction.
final_pareto_frontier.png	Margin versus worst harmonic residual, colored by total turns.
best_topology_designs/	Up to ten strong alternatives with different active-block topologies.
pareto_candidates.json	Certified selectable pool, radial archive, and search diagnostics.
campaign.json + run_manifest.json + inputs/	Exact settings, software versions, hashes and immutable conductor snapshots.

If nothing certifies

Do not immediately add generations. First inspect the near-feasible violation: geometry or current failures require a physically broader design space; a harmonic plateau may need more independent blocks or population; a small certification miss needs reserve.

PART III - RELIABILITY

Measured agreement with ROXIE in 1,000 blind air-core cases

DOT 1.0.0 was compared through a licensed ROXIE service at localhost:8080: 500 two-layer and 500 four-layer layouts, all generated without reference block coordinates.

Coverage

Aperture 20.004-39.968 mm; current 3.008-12.455 kA; 2-12 blocks; 5-44 turns.

Conductors

Eight Nb-Ti, Nb3Sn and mixed layer patterns from the validated catalogue.

Comparison

Identical R, turns, phi, alpha, current, conductor, symmetry and reference radius.

Quantity	Mean absolute	Median absolute	95th percentile	Maximum absolute	Signed mean
Bore field	0.0197%	0.0156%	0.0525%	0.0988%	+0.0181%
Peak field	0.6709%	0.6727%	1.1866%	2.1149%	+0.6235%
Load-line margin	0.1140 pp	0.1019 pp	0.2785 pp	0.6035 pp	-0.0243 pp
b3	1.2146 units	1.1119	2.7980	4.8190	+0.8080
b5	0.6507 units	0.5738	1.5252	2.4238	-0.0132
b7	0.2725 units	0.2314	0.7011	1.7254	-0.0502
b9	0.1035 units	0.0837	0.2610	0.5854	-0.0033
b11	0.0406 units	0.0351	0.1011	0.2547	+0.0053

Error definitions. Field errors are relative to ROXIE. Margin error is an absolute percentage-point difference. Harmonic error is an absolute accelerator-unit difference because relative error is not meaningful near zero.

What is strong

Bore-field parity is excellent. Peak field is close and DOT is conservative on average. Margin shows no material systematic bias.

What needs reserve

b3 is the most sensitive cross-tool result: 2.80 units at the 95th percentile and 4.82 units maximum.

Practical reliability rule

Within the sampled 2D coil-only domain, DOT is a reliable pre-design engine. For a 5-unit field-quality requirement, prefer a DOT worst harmonic near 2 units or less. Independently verify candidates close to acceptance. The parity study does not validate iron, ends, magnetization, mechanics, protection, tolerances or out-of-domain layouts.

Evidence: docs/validation/roxie_parity_1000_results.csv and roxie_parity_1000_summary.json. Deterministic seed 20260729; ROXIE runtime banner 23.6, git commit 26.1.0.b3.

PART IV - DESIGN CHECKLIST

A disciplined DOT campaign in ten checks

Use this page as the final design review before accepting a candidate for further engineering.

#	Check	Question answered
1	Scope	The task is a 2D, straight, coil-only dipole pre-design.
2	Cable data	Every layer uses the intended supported conductor and temperature.
3	Geometry convention	phi is zero at midplane; alpha is the absolute cable-frame angle.
4	Design space	Layer count, block/turn bounds and angle bounds do not exclude plausible solutions.
5	Clearances	Gaps and pole-turn proxy are physical values; numerical tolerance is only a classifier.
6	Current	Range or fixed-current mode is intentional and compatible with available turns.
7	Harmonics	Reference radius and signed targets represent the intended air-core specification.
8	Search	Quick is used for diagnosis; Design/Intensive and multiple seeds are used when warranted.
9	Certification	The chosen design is certified and has visible reserve in margin, harmonics, current and geometry.
10	Handoff	Campaign JSON, cadata snapshots, manifest, candidate JSON and geometry CSV remain together.

Ready-to-run example

GUI: choose campaign/dot_cables.cadata for every layer and select a supported conductor.
Command line:

```
dot validate campaign/7T_NbTi_noiron_sample.json
dot optimize campaign/7T_NbTi_noiron_sample.json --output results/7T_sample
```

Remember: a clean Pareto front is not a complete magnet. The next engineering steps must address iron, mechanical support and stress, protection, conductor degradation, tolerance robustness, end design, 3D effects and integration.

Primary references

K. Deb et al., "A fast and elitist multiobjective genetic algorithm: NSGA-II," IEEE Transactions on Evolutionary Computation 6(2), 2002, DOI 10.1109/4235.996017.
L. Bottura, "A practical fit for the critical surface of NbTi," IEEE Transactions on Applied Superconductivity 10(1), 2000.
DOT 1.1.0 source, tests, campaign archive and validation evidence dated 29 July 2026.